

Module 06: Learning in Robotics

Learning Objectives

1. Define **Learning** within the context of Robotics.
2. Analyze how Learning interacts with other components of the Active Inference framework.
3. Apply specific constraints of Robotics to the formal definition of Learning.

Introduction

This module explores **Learning**. In the **Robotics** curriculum, we approach this topic with a focus on specific applications and theoretical depth appropriate for the audience. Learning is a critical component of the 8-part Active Inference spine, bridging the gap between Action and Communication.

Key Concepts

1. Learning as a Markov Blanket Boundary

How does Learning define the boundary between the agent and the environment?

2. Generative Models of Learning

What parameters involved in Learning must be optimized to minimize variational free energy?

3. Active Inference Dynamics

How does the process of Learning drive the perception-action loop?

Applications

In Robotics, we see Learning manifest in: * **Specific Example 1**: A fly-inspired robot learns visuomotor associations using a spiking neural network architecture modeled on the

Drosophila mushroom body, where dopaminergic reward signals modulate synaptic weights between visual feature neurons and motor output neurons; this Hebbian-style learning updates the robot's generative model parameters so that over hundreds of trials, specific visual patterns become reliably associated with approach or avoidance motor programs, mirroring how fruit flies learn to associate odors with food or danger. * **Specific Example 2:** A snake-inspired hyper-redundant robot learns effective locomotion gaits (sidewinding, lateral undulation, concertina) for different terrain types by treating each surface interaction as evidence for updating the parameters of its body-terrain generative model; after traversing sand, gravel, and grass, the robot's learned precision matrices encode which gait pattern produces the least prediction error on each substrate, enabling it to automatically select the bio-inspired locomotion mode that best fits novel terrain without human specification.

Conclusion

Understanding Learning allows us to better model complex adaptive systems. In the next module, we will expand on this foundation.